

# MIT QUANTUM INFORMATION SCIENCE II — PART 3

## Course-Level QAI Learning Extraction — SU1–SU3

### Advanced Quantum Algorithms • Hidden Subgroup Algorithms • Quantum Information Theory

MIT Open Learning Library | Bhadale IT Hub | QAI Knowledge Capture | 2026

#### 1. COURSE-LEVEL IDENTITY & EVIDENCE SCOPE

- Course context: Quantum Information Science II, Part 3 — Advanced quantum algorithms and information theory.
- Consolidated subunits: SU1 Advanced Quantum Algorithms; SU2 Hidden Subgroup Algorithms; SU3 Quantum Information Theory.
- SU1 focuses on Abelian-group QFT, HSP, discrete logarithms, collision search, phase estimation and adaptive feedback-based factoring.
- SU2 focuses on Dihedral HSP, non-Abelian QFT, irreducible representations, Clebsch–Gordan decomposition, query/communication speedups, amplitude amplification, Hamiltonian simulation and entropy.
- SU3 focuses on Shannon and quantum information theory, typical sequences, source/channel coding, mutual information, channel capacity, quantum channels, random access codes, Holevo information and POVMs.
- Evidence boundary: MIT/course terminology, supplied questions, accepted graded results and displayed solutions are retained as source evidence. QAI Control Plane, VirtualQubit, registries, adaptive engines, information-aware orchestration and primitive/product mappings are architectural interpretations or research candidates, not MIT implementation claims.
- No independent executable notebook, simulator, hardware/QPU experiment or external benchmark was supplied in the three source packages.

#### 2. COURSE-LEVEL EXECUTIVE SYNTHESIS

- SU1 establishes how Fourier structure, hidden subgroups and eigenphase information can be converted into measurable evidence, with classical estimation and feedback controlling subsequent quantum execution.
- SU2 generalizes the mathematical foundation from Abelian to non-Abelian groups and adds representation theory, communication complexity, amplitude amplification, Hamiltonian simulation and entropy as reusable algorithmic/information primitives.
- SU3 establishes the information-theoretic boundary around computation: what information is represented, compressed, transmitted, correlated, measured and actually accessible from quantum states.
- The three units form a coherent QAI progression: hidden structure → mathematical representation → quantum transformation → measurement → classical inference → feedback/control → information/communication constraints.
- The strongest architectural lesson is that quantum execution should be treated as an evidence-producing component inside a hybrid computational workflow, with explicit metadata for mathematical assumptions, measurement records, confidence, information flow, resource constraints and validation.

#### 3. SU1 → SU3 LEARNING PROGRESSION

| Subunit | Primary Domain              | Key Mechanisms                                                                                                | QAI Contribution                                                                                         |
|---------|-----------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| SU1     | Advanced quantum algorithms | QFT, Abelian HSP, discrete log, Grover collision search, Kitaev phase estimation, adaptive feedback           | Adaptive Quantum Spectral Estimation, hidden-structure inference, convergence-aware hybrid control       |
| SU2     | Hidden subgroup algorithms  | Dihedral HSP, non-Abelian QFT, irreps, Clebsch–Gordan decomposition, amplification, simulation, communication | Representation-aware algorithm selection, sampling/inference orchestration, simulation/resource profiles |
| SU3     | Quantum information theory  | Entropy, typicality, coding, mutual information, channel capacity, density matrices, Holevo bound, POVMs      | Information-aware representation, communication, measurement planning and workload selection             |

#### 4. COURSE-WIDE CORE LEARNINGS

- Fourier analysis is a central mechanism for exposing hidden mathematical structure; group structure determines the appropriate Fourier representation.
- The algorithmic access model matters: standard input/output, oracle access and hidden-structure formulations lead to different algorithm choices and complexity assumptions.
- Representation theory becomes a practical computational resource in non-Abelian settings: irreps and tensor decompositions organize the information available after a quantum transform.
- Quantum sampling does not by itself produce the final answer; measurement records become evidence that classical inference reconstructs into hidden parameters, estimates or decisions.
- Adaptive execution closes a feedback loop: quantum probe → measurement → estimator update → changed quantum probe → convergence.
- Hybrid resource allocation is explicit in collision search: classical precomputation can be balanced against quantum search to optimize total query cost.
- Amplitude amplification is a reusable probability-boosting pattern rather than only a single search algorithm.

- Hamiltonian simulation requires explicit model, norm, approximation and resource assumptions.
- Entropy and mutual information provide mathematical tools for describing uncertainty, compression, correlation, information flow and the limits of reliable communication.
- Holevo information and POVMs establish an important boundary between information encoded in a quantum state and classical information that can actually be extracted through measurement.
- Information, communication capacity, measurement confidence and computational resources should be treated as managed execution metadata in a QAI system.
- VirtualQubit and Control Plane are best treated as abstraction/control layers that connect logical algorithms and information state to physical execution, without claiming a particular physical implementation.

## 5. COURSE-WIDE REUSABLE PATTERNS

| Pattern                                   | Reusable Structure                                                                                     | Source  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------|---------|
| Hidden-Structure Inference                | hidden function/group structure → Fourier/representation transform → measurement → classical inference | SU1/SU2 |
| Adaptive Quantum Estimation               | quantum operator/probe → interference → measurement → likelihood/estimator update → refined probe      | SU1     |
| Hybrid Search Allocation                  | classical precomputation → quantum search → optimize resource split                                    | SU1     |
| Representation-Aware Execution            | group/irrep metadata → decomposition → transform/measurement strategy → inference                      | SU2     |
| Quantum Sampling to Classical Inference   | repeated quantum samples → distribution estimate → reconstruction/verification                         | SU1/SU2 |
| Amplitude Boosting                        | prepare marked component → reflection/rotation → amplification → measurement                           | SU2     |
| Simulation-Abstraction                    | Hamiltonian model → representation/decomposition → approximate evolution → error/resource validation   | SU2     |
| Information-Aware Compression             | entropy/typicality → reduced representation → reconstruction                                           | SU3     |
| Channel-Aware Reliable Transfer           | source → encoder → noisy channel → decoder → reliability assessment                                    | SU3     |
| Information-Flow Analysis                 | X/Y/Z dependencies → mutual/conditional information → data-processing assessment                       | SU3     |
| Quantum-to-Classical Information Boundary | quantum state/ensemble → POVM/measurement → classical outcome → accessible information                 | SU3     |
| Evidence-Driven Hybrid Execution          | quantum transformation → measurement evidence → classical inference/control → validation               | SU1–SU3 |

## 6. QAI ARCHITECTURE MAPPING

- QAI Language — symbolic representation of groups, irreps, QFTs, HSP promises, operators, states, entropy, channels, measurements, POVMs, constraints and feedback policies.
- QAI Algorithm Layer — reusable QFT/HSP, hidden-structure, Grover/amplitude-amplification, phase-estimation, simulation, information-processing and measurement templates.
- QAI Compiler — representation-aware decomposition, tensor/irrep decomposition, circuit optimization, resource-aware mapping and information/measurement constraints.
- QAI Runtime — execute quantum transformations, controlled operators, measurements, sampling, communication and classical reconstruction.
- Hybrid Runtime — coordinate quantum state preparation/execution with classical estimation, inference, entropy/capacity calculations and adaptive feedback.
- QAI Control Plane — select algorithms/backends, manage shots and confidence, process measurement records, apply feedback policies, monitor channel conditions and route workloads.
- VirtualQubit — logical/virtual quantum-resource abstraction carrying state, mapping, measurement, provenance and lifecycle metadata; not a replacement for the physical qubit.
- Capability Registry — supported groups, irreps, QFT variants, controlled operations, amplification methods, simulation methods, measurement capabilities and backend constraints.
- Resource Registry — logical, virtual and physical quantum resources plus classical compute, memory, communication and measurement resources.
- Information Layer — entropy, mutual information, conditional information, channel profiles, density-matrix/state metadata and accessible-information constraints.
- Validation / Sanity Layer — normalization, probability, representation, unitarity, entropy, capacity, Holevo-bound, measurement and expected-distribution checks.
- QAI Labs — simulator-backed validation environment for QFT/HSP, phase estimation, adaptive convergence, representation theory, amplification, simulation and information-theoretic calculations.

## 7. COURSE-LEVEL QAI DATA / PRIMITIVE CANDIDATES

| Candidate                         | Purpose                                                                                          | Primary Source |
|-----------------------------------|--------------------------------------------------------------------------------------------------|----------------|
| QAI_GroupProfile                  | Group, order, generators, relations, Abelian/non-Abelian status and supported irreps             | SU2            |
| QAI_IrrepProfile                  | Representation label, dimension, matrices, equivalence and decomposition rules                   | SU2            |
| QAI_HSPProblem                    | Group, oracle, hidden subgroup promise, query model, measurement model and inference target      | SU1/SU2        |
| QAI_FourierTransformProfile       | Group, representation basis, normalization, forward/inverse transform and capability             | SU1/SU2        |
| AdaptiveQuantumSpectralEstimation | Estimate eigenphase/eigenparameter with measurement-driven feedback and convergence tracking     | SU1            |
| QAI_MeasurementRecord             | Basis, outcomes, shots, estimates, confidence and provenance                                     | SU1/SU2        |
| QAI_Constraint                    | Hidden parameters and equations/relations inferred from measurements                             | SU2            |
| QAI_SamplingProfile               | Target distribution, success probability, amplification and confidence                           | SU2            |
| QAI_SimulationProfile             | Hamiltonian representation, norm, approximation method, error budget and resources               | SU2            |
| InformationProfile                | Entropy, mutual information, conditional entropy and channel metrics                             | SU3            |
| TypicalSetReducer                 | Identify high-probability regions and compact representations                                    | SU3            |
| QAChannelProfile                  | Capacity, noise assumptions, rate and reliability targets                                        | SU3            |
| QuantumInformationState           | Density-matrix/state information and provenance                                                  | SU3            |
| POVMMeasurementPlanner            | Compare measurement strategies under accessible-information objectives                           | SU3            |
| HolevoBoundValidator              | Check proposed classical extraction against a quantum ensemble's Holevo bound                    | SU3            |
| InformationAwareWorkloadRouter    | Use information characteristics plus execution capability for classical/quantum/hybrid placement | SU3            |

## 8. COURSE-LEVEL INFORMATION / EXECUTION FLOW

1. Classify the mathematical/problem model: group structure, oracle/promise, information source, communication model and target output.
2. Select an appropriate representation: Abelian/non-Abelian Fourier basis, irreps, density matrix, probability distribution or other validated model.
3. Construct the quantum algorithm/probe and identify its measurement and inference requirements.
4. Estimate resource requirements: queries, shots, communication, circuit depth, classical processing, memory and channel constraints.
5. Execute the quantum component and capture measurement evidence with provenance.
6. Use classical inference, reconstruction, entropy/capacity analysis or decision logic to interpret the evidence.
7. If the algorithm is adaptive, update the estimator/model and select the next quantum operation from the feedback policy.
8. Check confidence, convergence, normalization, information bounds, capacity assumptions and other applicable validation conditions.
9. Select or adapt classical, quantum or hybrid execution according to evidence and policy.
10. Capture final result, execution trace and validation evidence for reuse or independent review.

## 9. COURSE-LEVEL VALIDATION / ASSURANCE MODEL

| Dimension          | SU1                                                                                       | SU2                                                                                                | SU3                                                                                                |
|--------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Primary evidence   | QFT/HSP derivations, collision optimization, phase distributions and adaptive convergence | Irreps, Dihedral HSP, QFT measurement, amplification, simulation, communication and entropy        | Entropy, typicality, coding, channel capacity, mutual information, quantum states and Holevo/POVM  |
| Key failure signal | Incorrect phase/estimate, unstable convergence, resource or feedback mismatch             | Invalid representation, normalization/probability error, insufficient sampling or simulation error | Entropy/capacity/Holevo inconsistency, measurement probability error or information-flow violation |
| Response           | Adapt probe/ $\theta$ , increase repetitions, retry or change strategy                    | Change representation/decomposition, increase sampling, validate or rebind backend                 | Change representation/encoding/channel/measurement strategy or escalate                            |
| Validation         | Simulator, phase distributions, convergence, resource scaling and later QPU tests         | Representation/state checks, algorithm simulators, communication and simulation error tests        | Entropy, typicality, capacity, measurement, POVM and Holevo-bound checks                           |

## 10. CROSS-SUBUNIT CONVERGENCE

| Convergent Theme             | SU1 → SU2 → SU3                                                                                          | QAI Interpretation                                                          |
|------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Hidden information           | HSP/phase information → non-Abelian hidden subgroup structure → information encoded vs accessible        | Represent hidden state/structure explicitly and expose evidence boundaries  |
| Mathematical representation  | QFT/characters → irreps/Clebsch–Gordan → probability/density-matrix/information measures                 | Representation choice becomes a first-class compiler/control-plane decision |
| Quantum → classical boundary | Phase measurement → HSP sampling → POVM/classical outcome                                                | Measurement is an evidence interface, not merely a terminal operation       |
| Hybrid execution             | Phase estimation + classical estimation → sampling + inference → coding/measurement + classical decoding | Quantum and classical components should be orchestrated as one workflow     |
| Feedback                     | Adaptive $\theta$ → confidence/sampling management → information-aware routing                           | Runtime telemetry can drive the next computational action                   |
| Resource awareness           | Queries/repetitions → communication/simulation resources → capacity/compression limits                   | Execution decisions should include resource and information constraints     |

## 11. COURSE-LEVEL ARCHITECTURAL PRINCIPLES

- Represent mathematical assumptions explicitly before execution.
- Choose representation and algorithm based on structure rather than treating all quantum workloads uniformly.
- Treat measurement records as first-class evidence with provenance, confidence and sampling metadata.
- Separate quantum state/information from classical information that can actually be extracted.
- Use adaptive feedback when the measurement record can improve subsequent quantum control.
- Treat communication capacity, entropy, information flow and accessible information as execution constraints where applicable.
- Keep logical/virtual state abstractions distinct from physical realization.
- Use deterministic validation and policy rules for safety-critical decisions; optimization/AI can select among validated strategies.
- Promote candidates only after independent implementation, simulator validation, recurrence and hardware validation where relevant.
- Do not infer production maturity, hardware scalability, security certification or quantum advantage from the educational evidence alone.

## 12. VALIDATION & PROMOTION ROADMAP

1. Independently reproduce the supplied QFT, HSP, representation, entropy, coding, measurement and Holevo calculations.
2. Build simulator-backed reference implementations for QFT/HSP, phase estimation/AQSE, Dihedral HSP, amplitude amplification and selected Hamiltonian-simulation examples.
3. Validate state normalization, probabilities, representation dimensions, unitary assumptions and expected distributions.
4. Inject controlled noise and insufficient sampling to test confidence, convergence, retry and reject/escalate behavior.
5. Benchmark adaptive phase estimation against non-adaptive variants using error, shots, iterations, circuit depth and classical estimator cost.
6. Validate communication assumptions and resource costs for distributed algorithms.
7. Validate entropy, typical-set, channel-capacity, random-access-code and Holevo/POVM examples independently.
8. Verify cross-layer consistency between logical intent, VirtualQubit metadata, measurement records, backend mapping and telemetry.
9. Perform IV&V for governed capabilities and record requirements, assumptions, evidence and acceptance criteria.
10. Promote only recurring, independently validated primitives and templates into the QAI Enterprise Library / Product Foundry.

## 13. COURSE MATURITY & EVIDENCE ASSESSMENT

- Educational knowledge: strong for the supplied SU1–SU3 concept and problem-set evidence.
- Mathematical foundation: strong across Fourier analysis, representation theory, hidden-subgroup algorithms, estimation, entropy and quantum information.
- Algorithmic foundation: strong for the supplied QFT/HSP, collision search, phase estimation, amplitude amplification and information-theoretic constructions.
- QAI architectural convergence: strong — adaptive hybrid execution, representation-aware control and information-aware orchestration recur across all three units.
- Implementation evidence: the supplied records are based on MIT OLL screenshots, accepted results and displayed solutions; no independent implementation was inspected.
- Reusable capabilities: candidates only until independent implementation, benchmarks and validation are completed.
- Production maturity: not established by these educational records.

## 14. COURSE RETENTION SUMMARY

- SU1 should be retained as the foundation for Fourier-based hidden-structure inference, phase estimation and adaptive quantum-classical feedback.
- SU2 should be retained as the foundation for non-Abelian representation-aware algorithms, Dihedral HSP, amplification, Hamiltonian simulation, communication complexity and entropy diagnostics.

- SU3 should be retained as the information-theoretic foundation for compression, communication, information flow, quantum-state characterization, measurement and accessible-information limits.
- Together, Part 3 provides a strong conceptual bridge from quantum algorithms to mathematical structure and then to the information limits governing what can be represented, transmitted and extracted.
- Master pattern: Structure → Representation → Quantum Transformation → Measurement → Evidence → Classical Inference → Feedback/Control → Information/Resource Validation.

## 15. PROVENANCE & FIDELITY NOTE

This course-level section is synthesized from the three supplied SU extraction records. The source packages' terminology, framing, accepted evidence and limitations are retained. QAI architecture, primitive and product mappings are explicitly study-session interpretations or research/design candidates and should not be read as MIT-prescribed implementations.

# PART II — COMPLETE SU1–SU3 SOURCE EXTRACTIONS

Original supplied extraction records appended for traceability and future course-level reuse.

# SU1 — Advanced Quantum Algorithms

Complete supplied extraction record appended without altering its evidence boundary.

# SU2 — Hidden Subgroup Algorithms

Complete supplied extraction record appended without altering its evidence boundary.

# SU3 — Quantum Information Theory

Complete supplied extraction record appended without altering its evidence boundary.

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## Quantum Information Science II, Part 3 SU1 — Advanced Quantum Algorithms

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### Standard Course Extraction Template — Completed Subunit / Evidence Package

Evidence boundary: MIT OLL/course learning is recorded from the supplied study-session screenshots, question text, accepted answers and displayed official solutions. QAI architecture mappings, primitive candidates and product implications are study-session interpretations/design candidates, not claims that MIT specifies or validates a production QAI implementation.

**Extraction date:** 2026-08-17

**Extraction status:** COMPLETED for the SU1 evidence reviewed in this study session

**Evidence source:** User-provided MIT Open Learning Library screenshots, problem-set questions, accepted graded results, displayed solutions and study-session reasoning

**Notebook / executable asset:** None supplied or inspected for this SU1 package

## 1. Course Identity

| Field                         | Captured information                                                                                                                                                                                                         |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Course title                  | Quantum Information Science II, Part 3 — Advanced quantum algorithms and information theory                                                                                                                                  |
| MITx / course ID              | Not explicitly supplied in the current screenshots; course context is retained from the study session as the Part 3 advanced-algorithms course.                                                                              |
| Term / course version         | Not supplied in the inspected evidence package.                                                                                                                                                                              |
| MIT Open Learning Library URL | Not supplied in the inspected evidence package.                                                                                                                                                                              |
| Primary domain                | Advanced quantum algorithms, quantum Fourier transforms over groups, hidden subgroup problems, discrete logarithms, hidden structures, quantum search/collision algorithms, phase estimation and adaptive quantum factoring. |
| Subunit                       | SU1 — Advanced quantum algorithms                                                                                                                                                                                            |
| Purpose                       | Capture the supplied SU1 lecture/problem-set evidence and extract reusable mathematical, algorithmic, estimation, feedback and QAI architecture patterns.                                                                    |
| Source material available     | User-provided MIT OLL screenshots, question text, accepted graded answers, displayed official solutions and study-session discussion.                                                                                        |
| Extraction status             | Completed for the SU1 evidence reviewed; full independent page-by-page course crawl was not performed.                                                                                                                       |

## 2. Course Fidelity and Source Structure

- MIT terminology and the course's mathematical framing are preserved at the source-evidence layer.
- Lecture/concept questions and problem-set evidence are kept separate.
- Solved questions are treated as first-class evidence; visible green checks / accepted scores were used as confirmation.
- Some ungraded feedback/convergence exercises displayed greyed-out submission controls and official solutions; these are retained as course evidence rather than treated as graded failures.
- A parser/input issue occurred in at least one multi-part feedback-repetition item; the displayed course solution was used to resolve the mathematical content.
- No executable notebook, independent simulator, hardware/QPU experiment or external benchmark was supplied for this SU1 package.
- QAI architecture interpretations are explicitly separated from MIT evidence.

## 3. Course / Subunit Map

| Item                                                           | Type              | Evidence / result                                        | Status    |
|----------------------------------------------------------------|-------------------|----------------------------------------------------------|-----------|
| SU1 — Advanced quantum algorithms                              | Subunit           | Supplied screenshots and study-session evidence          | Completed |
| Quantum Fourier transform over an Abelian group                | Lecture / concept | 1/1 graded                                               | Completed |
| Models of quantum algorithms                                   | Lecture / concept | 1/1 after correcting option interpretation               | Completed |
| Abelian quantum Fourier transform                              | Lecture / concept | 1/1 graded                                               | Completed |
| Quantum Fourier transform over $\mathbb{Z}_N$                  | Lecture / concept | 1/1 graded                                               | Completed |
| Hidden subgroup problem (HSP)                                  | Lecture / concept | Reviewed; accepted selections identified                 | Completed |
| Abelian HSP                                                    | Lecture / concept | 1/1 graded                                               | Completed |
| Discrete log                                                   | Lecture / concept | 1/1 graded                                               | Completed |
| Non-Abelian HSP                                                | Lecture / concept | 1/1 graded                                               | Completed |
| Generalized Bernstein–Vazirani: learning linear function mod 3 | Problem set       | Classical lower bound, math interlude, quantum algorithm | Completed |
| Quantum algorithm to break hash functions                      | Problem set       | Collision search and K optimization                      | Completed |
| Quantum algorithm for computing the discrete log               | Problem set       | Multi-part Fourier/group derivation                      | Completed |
| Hidden Parabola                                                | Problem set       | Multiple derivation/measurement parts                    | Completed |
| Quantum factoring as a feedback process                        | Problem set       | Kitaev-style eigenphase estimation and repetition        | Completed |
| Quantum factoring by feedback: convergence                     | Problem set       | Adaptive estimator and feedback-control derivation       | Completed |

## 4. Module / Subunit Record

**Subunit ID / name:** SU1 — Advanced quantum algorithms

**MIT source URL:** Not supplied in the inspected evidence package.

**Learning objectives:** Understand QFT over Abelian groups; distinguish standard input-output, oracle and hidden-structure models; understand Abelian and non-Abelian HSP framing; connect Fourier analysis to hidden structure; understand discrete-log and collision problems; derive phase/eigenvalue estimation and feedback-based factoring.

**Lectures / topics:** Abelian-group QFT; models of quantum algorithms; characters and roots of unity; QFT over  $\mathbb{Z}_N$ ; HSP; Abelian HSP; discrete logarithm; non-Abelian HSP.

**Problem sets:** Generalized Bernstein–Vazirani mod 3; quantum collision/hash search; discrete logarithm; Hidden Parabola; Kitaev quantum factoring by feedback and convergence.

**Important concepts:** Characters, roots of unity, Abelian groups, tensor-product QFT, uncomputing, HSP, subgroup encoding, linear equations, discrete logarithm, Grover collision search, eigenstates/eigenphases, phase kickback, Bayesian/conditional estimation, adaptive feedback and convergence.

**QAI interpretation focus:** Algorithm-model selection, mathematical validity/intake checks, quantum/classical hybrid execution, adaptive estimation, feedback control, measurement telemetry and convergence-aware workflow execution.

## 5. Lecture / Content Record

| Lecture / item                                  | Type             | Core concepts                                                                                                                                  | Key QAI relevance                                              |
|-------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Quantum Fourier transform over an Abelian group | Concept question | Characters, group multiplication, QFT; phase estimation uses conventional QFT framing.                                                         | Group-aware transform capability and representation selection. |
| Models of quantum algorithms                    | Concept question | Standard input-output model, quantum oracle model, Shor as standard-model example, phase estimation as oracle-model example.                   | Model/algorithm classifier before execution.                   |
| Abelian quantum Fourier transform               | Concept question | Character maps, roots of unity, $\mathbb{Z}_N$ as Abelian group, character evaluation.                                                         | Mathematical metadata and group capability registry.           |
| Quantum Fourier transform over $\mathbb{Z}_N$   | Concept question | QFT from phase estimation; uncomputing; product-group tensor decomposition.                                                                    | Transform composition and dependency tracking.                 |
| Hidden subgroup problem                         | Concept question | Hidden subgroup encoding, non-standard information-access model, Simon as HSP example.                                                         | Hidden-structure solver abstraction.                           |
| Abelian HSP                                     | Concept question | Linear-equation solution, character values and subgroup-dependent output distribution.                                                         | Constraint extraction and inference service.                   |
| Discrete log                                    | Concept question | Discrete log as HSP over $\mathbb{Z}_N \times \mathbb{Z}_N$ ; group-specific rather than arbitrary-group formulation in the supplied material. | Group capability matching and solver selection.                |
| Non-Abelian HSP                                 | Concept question | Gentle measurement conditions, Heisenberg group, triangular matrices, non-Abelian structure.                                                   | Advanced group/model metadata and measurement constraints.     |

## 6. Problem Set Record

| Problem set / item                               | Question focus                                                                                                 | Source evidence                                                           | Status                        |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------|
| Generalized Bernstein–Vazirani                   | Classical lower bound for learning a linear function mod 3; roots-of-unity identity; quantum learning circuit. | User screenshots; accepted answers.                                       | Completed                     |
| Quantum algorithm to break hash functions        | Two-to-one collision search using K classical queries plus Grover search; optimize K.                          | User screenshots; accepted answers.                                       | Completed                     |
| Quantum algorithm for computing the discrete log | HSP/QFT derivation, Fourier amplitudes and extraction of discrete-log information.                             | User screenshots; accepted multi-part result.                             | Completed                     |
| Hidden Parabola                                  | Inverse QFT, two copies, measurement, quadratic phase and analogue of coset states.                            | User screenshots; accepted multi-part results.                            | Completed                     |
| Quantum factoring by feedback: repetition        | Kitaev feedback repetition; superposition of eigenstates; conditional distribution $p(k n)$ .                  | User screenshots; some parser/input issue in one entry; official solution | Completed / solution captured |

|                                            |                                                                                                                        |                                                 |                               |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|-------------------------------|
|                                            |                                                                                                                        | displayed for ungraded parts.                   |                               |
| Quantum factoring by feedback: convergence | Adaptive estimator $\theta$ , measurement-record update, $\theta$ -box feedback circuit and convergence toward $k/r$ . | User screenshots; official solutions displayed. | Completed / solution captured |

## 7. Solved Question Extraction

### 7.1 Generalized Bernstein–Vazirani — classical lower bound

**Problem statement:** Learn unknown  $s \in \mathbb{Z}_3^n$  from black-box  $f(x) = x \cdot s \bmod 3$ .

**Required result:**  $n$  classical queries are required to determine  $s$  with certainty.

**Solution / reasoning:** Query basis vectors / independent inputs to recover each coefficient.

**Key insight:** The quantum algorithm is motivated by a linear-function learning problem whose classical query lower bound is  $n$ .

**Evidence / limitation:** Educational mathematical/algorithm evidence; no independent implementation supplied.

### 7.2 Learning a linear function mod 3 — math interlude

**Problem statement:** Find  $z \in \mathbb{C}$  such that  $\sum_v z^{\langle u, v \rangle}$  equals  $3^n$  for  $u=0$  and 0 otherwise.

**Required result:**  $z = \exp(2\pi i/3)$ .

**Solution / reasoning:** Root-of-unity cancellation makes nonzero Fourier characters sum to zero.

**Key insight:** Roots of unity provide the interference mechanism behind the mod-3 Fourier transform.

**Evidence / limitation:** Directly supports QFT/character-based inference.

### 7.3 Learning a linear function mod 3 — quantum algorithm

**Problem statement:** Use  $U_f|x\rangle|y\rangle = |x\rangle|y+s \cdot x \bmod 3\rangle$  and the appropriate qutrit input state.

**Required result:** The accepted input state is  $(|00\rangle + \exp(2\pi i/3)|01\rangle + \exp(4\pi i/3)|10\rangle)/\sqrt{3}$  in the course's binary-labelled qutrit notation.

**Solution / reasoning:** Prepare Fourier/character state, query  $U_f$ , apply inverse Fourier processing and measure to recover  $s$ .

**Key insight:** Quantum interference extracts the hidden linear coefficients without the classical  $n$ -query pattern.

**Evidence / limitation:** Parser-safe symbolic representation was accepted in the supplied screenshot.

### 7.4 Quantum algorithm to break hash functions — collision search

**Problem statement:** Find  $x_1 \neq x_2$  with  $f(x_1) = f(x_2)$  for a two-to-one black-box  $f: [N] \rightarrow S$ .

**Required result:** Query  $K$  values, then use Grover to find a new input colliding with one of the stored values.

**Solution / reasoning:** Query complexity  $K + \sqrt{N/K}$ .

**Key insight:** Balancing classical precomputation against quantum search yields the optimal trade-off.

**Evidence / limitation:** Supports hybrid classical-memory + QPU-search workflows.

### 7.5 Optimizing $K$

**Problem statement:** Minimize  $K + \sqrt{N/K}$  by choosing  $K$  proportional to  $N^a$ .

**Required result:**  $a=1/3$  and resulting query exponent  $b=1/3$ .

**Solution / reasoning:** Balance the two cost terms asymptotically.

**Key insight:** Hybrid algorithms can expose an explicit classical/quantum resource allocation parameter.

**Evidence / limitation:** Candidate resource-aware scheduler input.

## 7.6 Discrete logarithm — HSP formulation

**Problem statement:** Use an HSP over  $\mathbb{Z}_N \times \mathbb{Z}_N$  to encode the discrete logarithm.

**Required result:** The supplied lecture evidence states discrete log can be solved using HSP on  $\mathbb{Z}_N \times \mathbb{Z}_N$ .

**Solution / reasoning:** Apply Fourier/group reasoning and recover the hidden subgroup relation.

**Key insight:** Hidden-structure formulation changes the computational access model.

**Evidence / limitation:** Group metadata becomes an algorithm-selection dependency.

## 7.7 Hidden Parabola

**Problem statement:** Recover hidden quadratic parameters  $\alpha$  and  $\beta$  from a black-box function whose level sets are parabolas.

**Required result:** Use Fourier transforms and repeated copies; measured states encode quadratic phase information.

**Solution / reasoning:** Quadratic phase terms are transformed into measurable linear relations; repeated copies improve extraction.

**Key insight:** A structured hidden function can be converted into a phase-estimation / interference problem.

**Evidence / limitation:** Potential pattern for structured inverse problems.

## 7.8 Quantum factoring by feedback — eigenstate formulation

**Problem statement:** Estimate the eigenphase of  $U|m\rangle = |ym \bmod N\rangle$  where eigenphases are  $e^{-2\pi i \phi_k}$ ,  $\phi_k = k/r$ .

**Required result:** Kitaev-style ancilla circuit gives  $p_0 = \cos^2(\pi\phi)$  for a one-cycle eigenphase probe.

**Solution / reasoning:** Hadamard  $\rightarrow$  controlled- $U \rightarrow$  Hadamard  $\rightarrow$  ancilla measurement.

**Key insight:** Factoring is viewed as hidden-order/eigenphase estimation rather than only as a fixed circuit.

**Evidence / limitation:** Core evidence for adaptive hybrid quantum-classical execution.

## 7.9 Quantum factoring by feedback — repetition

**Problem statement:** Start from  $|1\rangle$ , an equal superposition of eigenstates, and repeat the phase-probe experiment  $t$  times.

**Required result:**  $p(k,n) = (1/r) C(t,n) \cos^{2n}(\pi k/r) \sin^{2(t-n)}(\pi k/r)$ .

**Solution / reasoning:** Condition on the number  $n$  of zero outcomes; isolate  $k$ -dependent likelihood terms.

**Key insight:** Increasing  $t$  sharpens the conditional distribution around informative  $k$  values.

**Evidence / limitation:** Peak width scales as  $O(1/t)$  in the displayed course derivation.

## 7.10 Quantum factoring by feedback — numerical example

**Problem statement:** Use  $N=143$ ,  $y=5$ ,  $r=20$  as a numerical illustration.

**Required result:** Peaks occur around  $k=6$  and  $k=14$ ; the inferred order is  $\approx 20$ .

**Solution / reasoning:** Verify  $5^{20} \bmod 143 = 1$ ; then  $5^{10} \bmod 143 = 12$  and factors are  $12 \pm 1 = 13, 11$ .

**Key insight:** Quantum phase information is followed by classical modular verification and factor extraction.

**Evidence / limitation:** Strong example of quantum measurement → classical validation.

## 7.11 Quantum factoring by feedback — convergence

**Problem statement:** Improve the unstable early estimator by updating  $\theta$  from the measurement record.

**Required result:** If  $b_n=0$ ,  $f_{n+1}=\cos^2(\phi-\theta)f_n$ ; if  $b_n=1$ ,  $f_{n+1}=\sin^2(\phi-\theta)f_n$ ; choose  $\theta$  maximizing  $f_{n+1}$ .

**Solution / reasoning:** Run cycle → record  $b_n$  → update model → choose  $\theta$  → repeat.

**Key insight:** The experiment itself becomes adaptive: measurement history changes the next quantum operation.

**Evidence / limitation:** Course solution states asymptotic  $\theta$  approaches  $k/r$  with  $O(\log N)$  iterations.

## 8. Notebook / Computational Asset Record

| Field                             | Captured information                                                                                         |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------|
| Notebook name                     | None supplied or inspected.                                                                                  |
| Course / problem-set relationship | No executable notebook evidence supplied; several exercises were numerical/algorithmic in nature.            |
| Original source                   | MIT OLL screenshots and displayed course solutions.                                                          |
| Language                          | Not applicable.                                                                                              |
| Framework / SDK                   | Not supplied.                                                                                                |
| Dependencies                      | Not supplied.                                                                                                |
| Input                             | Problem-specific mathematical parameters such as $N$ , $y$ , $r$ , $t$ , $k$ and group/function definitions. |
| Output                            | Analytical expressions, probabilities, estimated phases/orders and classical verification results.           |
| Algorithm / model                 | QFT/HSP reasoning, Grover collision search, phase estimation and adaptive feedback.                          |
| Execution environment             | MIT OLL learning environment; no independent runtime evidence.                                               |
| Result                            | Displayed answers/solutions and accepted graded results.                                                     |
| Validation                        | Course grading and displayed official solutions; independent implementation remains required.                |

## 9. Algorithm / Mathematical Model Record

| Model / algorithm            | Core mathematical structure                                                                  | Inputs                                                      | Outputs                                                 | Complexity / evidence                                                               |
|------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------|
| QFT over Abelian group       | Character-based Fourier transform; roots of unity for cyclic groups.                         | Finite Abelian group / quantum register.                    | Phase/amplitude representation in Fourier basis.        | Educational derivation; no independent implementation.                              |
| HSP                          | Hidden subgroup encoded through an oracle/group-access model.                                | Group, subgroup promise, oracle/function access.            | Information identifying hidden subgroup.                | Model depends on oracle/group access assumptions.                                   |
| Discrete log via HSP         | HSP on $\mathbb{Z}_N \times \mathbb{Z}_N$ .                                                  | Group element / generator relation.                         | Discrete logarithm information.                         | Educational algorithmic evidence.                                                   |
| Grover collision search      | $K$ classical queries + Grover search; total $K+\sqrt{(N/K)}$ .                              | Two-to-one black-box $f$ and stored $K$ outputs.            | Collision pair.                                         | Optimization gives $K \propto N^{1/3}$ , query exponent $1/3$ .                     |
| Kitaev phase estimation      | Eigenphase $\phi_k=k/r$ exposed through controlled- $U$ and ancilla interference.            | Unitary $U$ , eigenstate/superposition, repetitions.        | Eigenphase/order estimate.                              | Repetition peak width $O(1/t)$ ; adaptive convergence shown conceptually.           |
| Adaptive feedback estimation | Likelihood model updated from measurement record; $\theta$ chosen to maximize updated model. | Measurement history, current $\theta$ , controlled- $U^x$ . | Updated $\theta$ , phase estimate, convergence history. | Course solution states asymptotic $O(\log N)$ iteration behavior for the estimator. |

## 10. Extracted Pattern

**Pattern name:** Adaptive Quantum Spectral Estimation & Feedback

**Source course / module:** Quantum Information Science II, Part 3 — SU1 Advanced Quantum Algorithms; strongest evidence from quantum factoring by feedback and convergence.

**Source problems:** Quantum factoring by feedback: repetition; repetition IV/V; convergence II/III; related phase-estimation and HSP questions.

**Problem-solving structure:** Quantum operator → quantum probe → interference / phase kickback → measurement → statistical estimation → adaptive parameter update → refined quantum probe.

**Inputs:** Quantum operator  $U$ ; state preparation; measurement configuration; estimator state; feedback policy; resource/execution context.

**Outputs:** Estimated phase/eigenparameter; confidence/uncertainty; measurement history; convergence state; next control parameters.

**Control / feedback:** Measurement results update the likelihood/model;  $\theta$  is selected to maximize the updated model; controlled- $U^x$  and  $\theta$ -box parameters are changed through classical feedback.

**Recurring evidence:** QFT/HSP exposes hidden structure through Fourier information; phase estimation converts eigenphase into measurable data; factoring uses classical post-processing; convergence introduces adaptive feedback.

**Potential reuse:** Phase estimation, quantum sensing, device characterization, Hamiltonian learning, adaptive quantum algorithms, quantum control, simulation, optimization and QAI digital-twin synchronization.

**Confidence / evidence level:** High for the educational algorithmic pattern; medium for broader QAI reuse because no independent implementation or hardware evidence was supplied.

## 11. Primitive Candidate

**Primitive name:** Adaptive Quantum Spectral Estimation (AQSE)

**Source evidence:** Kitaev-style quantum factoring by feedback, eigenphase estimation, repetition/convergence and adaptive  $\theta$ -controller material.

**Pattern:** Quantum Operator → Probe → Measurement → Estimation → Feedback → Refined Probe.

**Purpose:** Estimate hidden spectral/eigenphase parameters of a quantum operator and adapt subsequent execution until a configured precision/confidence condition is reached.

**Proposed inputs:** operator; state\_preparation; estimation\_config; feedback\_policy; execution\_context.

**Proposed outputs:** estimated\_phase; eigenvalue/eigenparameter; confidence interval; uncertainty; observations; convergence history; iterations; resource usage; execution trace; status.

**API concept:** AdaptiveQuantumSpectralEstimation(operator, state\_preparation, estimation\_config, feedback\_policy, execution\_context)

**Runtime requirements:** Controlled quantum operations, ancilla preparation/measurement, coherent execution, measurement telemetry, classical estimation, feedback control and stopping/retry policy.

**Dependencies:** QAI Runtime; Hybrid Runtime; measurement/telemetry service; mathematical estimation service; capability/resource registries; backend execution interface.

**Validation required:** Independent simulator implementation; multiple phase-estimation examples; noisy simulations; resource scaling; adaptive-control stability; hardware/QPU validation; comparison with non-adaptive QPE/Kitaev variants.

**Maturity:** Candidate / research-design level; no production implementation evidence in this SU1 package.

**Promotion recommendation:** Retain as a candidate reusable QAI primitive. Promote only after independent reference implementation and cross-course validation.

## 12. QAI Architecture Mapping

| QAI layer               | Proposed mapping                                                                                                                           | Evidence classification             |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| QAI Language            | Represent QFT, HSP, characters, group operations, quantum probes, phase estimates, measurement records and feedback policies symbolically. | QAI interpretation                  |
| QAI Primitives          | AQSE candidate; QFT/HSP/phase-estimation supporting primitives; Grover collision-search capability.                                        | Candidate                           |
| QAI OS                  | Manage hybrid workflow lifecycle, estimator state, retries, stopping conditions and resource-aware execution.                              | Proposed architecture               |
| QAI Runtime             | Execute QFT, controlled-U, measurement, group operations and classical reconstruction.                                                     | Conceptual mapping                  |
| Hybrid Runtime          | Coordinate QPU/QPU-simulator execution with classical estimation, probability calculations and validation.                                 | Strong conceptual fit               |
| QAI Control Plane       | Select $\theta$ , $x$ , repetition count, measurement strategy, backend and stopping policy from telemetry.                                | Strong architectural interpretation |
| Quantum Control Plane   | Translate logical probe/control requests to backend-specific controlled operations and measurements.                                       | Proposed                            |
| Adaptive Network Fabric | Potentially transport feedback metadata and resource/capability information in distributed QAI workflows.                                  | No direct SU1 evidence              |
| Resource Registry       | Track QPU/simulator capabilities, qubit counts, measurement constraints, runtime and resource cost.                                        | Proposed                            |
| Capability Registry     | Register QFT variants, HSP solvers, controlled-U, phase-estimation, Grover and adaptive-estimation capabilities.                           | Proposed                            |
| QAI Labs                | Natural environment for simulator-backed validation of QFT/HSP/QPE/AQSE and convergence behavior.                                          | Strong fit                          |
| QAI Product Foundry     | Potential future productization path after independent validation and systems engineering.                                                 | Future / candidate                  |

## 13. Classification

| Category                | Assessment                                                                                                                     |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Educational knowledge   | Yes — primary classification.                                                                                                  |
| Research                | Yes — adaptive estimation and hidden-structure patterns are research/design inputs; no new research result established here.   |
| Mathematical model      | Strong — characters, roots of unity, Fourier transforms, HSP, eigenphases, conditional distributions and feedback likelihoods. |
| Algorithm               | Strong — QFT/HSP, collision search, discrete-log formulation, phase estimation and adaptive factoring.                         |
| Pattern                 | Strong candidate — Probe → Measure → Estimate → Adapt.                                                                         |
| Prototype               | Not demonstrated in this evidence package.                                                                                     |
| Implementation evidence | None independent of the MIT OLL learning environment.                                                                          |
| Reusable capability     | Candidate — AQSE and supporting QFT/HSP capabilities require validation.                                                       |
| Future capability       | Yes — adaptive spectral estimation service, hybrid feedback engine and convergence-aware execution control.                    |

## 14. Evidence and Provenance

**MIT OLL source URL(s):** Not supplied in the inspected evidence package.

**Original document / problem-set reference:** User-provided screenshots from SU1 Advanced Quantum Algorithms; displayed problem-set solutions and accepted graded results.

**Notebook reference:** None supplied.

**Solution reference:** Official solutions displayed within the MIT OLL problem pages for the ungraded feedback/repetition and convergence exercises; accepted answers for graded items.

**External reference (if any):** No external web source was used as primary evidence for this extraction.

**Evidence limitations:** Screenshot-based extraction; exact full MIT page hierarchy, official URLs, complete question inventory, executable notebooks, simulator results and QPU results were not supplied. Some course entry fields exposed parser/input limitations. QAI mappings are interpretations and must not be read as MIT implementation claims.

## 15. Cross-Course Convergence

| Course / evidence                          | Topic / problem                    | Common pattern                                          | Primitive candidate                         | Evidence                                   | Status              |
|--------------------------------------------|------------------------------------|---------------------------------------------------------|---------------------------------------------|--------------------------------------------|---------------------|
| Earlier quantum algorithms course evidence | QFT, QPE, Shor/order finding       | Phase encoding → measurement → classical reconstruction | Phase Estimation / Quantum Phase Extraction | Existing extraction record in File Library | Initial convergence |
| Current SU1                                | HSP, discrete log, hidden parabola | Hidden structure → Fourier representation → inference   | Hidden Structure / Fourier Inference        | Current screenshots                        | Candidate           |
| Current SU1                                | Quantum factoring by feedback      | Probe → Measure → Estimate → Adapt                      | AQSE                                        | Current screenshots + displayed solutions  | Strong candidate    |
| Current SU1                                | Grover collision search            | Classical precomputation + quantum search trade-off     | Hybrid Search / Collision Optimizer         | Accepted problem-set result                | Candidate           |
| QAI architecture work                      | Hybrid Runtime / Control Plane     | Quantum observation drives classical control            | Adaptive Quantum Loop                       | Study-session design interpretation        | Research candidate  |

Cross-course convergence is an architectural observation and does not replace independent validation.

## 16. Reuse and Implementation Assessment

- Direct reuse: mathematical definitions of QFT, characters, roots of unity, HSP and phase-estimation models as reference knowledge.
- Pattern reuse: strong candidate for adaptive measurement/estimation/control workflows.
- Design reference: estimator state, measurement history, convergence metrics and feedback parameters can become explicit QAI workflow metadata.
- Research input: adaptive quantum spectral estimation, hidden-structure inference and convergence-aware execution.
- Architecture input: QAI Runtime, Hybrid Runtime, QAI Control Plane, Capability Registry and Resource Registry.
- Prototype candidate: simulator-backed AQSE reference implementation.
- New implementation required: yes — no production implementation evidence was supplied.
- Historical / educational reference only: individual MIT exercises remain educational evidence until independently implemented and validated.

## 17. Validation Plan

| Validation area                 | Required evidence                                                                                                                                      |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reference implementation needed | Python simulator implementation of QFT/HSP/phase-estimation/AQSE loops with reproducible test vectors.                                                 |
| Simulator validation            | Verify roots-of-unity identities, QFT transforms, HSP toy instances, phase distributions, repetition peak narrowing and adaptive $\theta$ convergence. |
| Hardware / QPU validation       | Run small controlled-U / phase-estimation experiments; compare measured distributions with simulator predictions under device noise.                   |
| Independent problem validation  | Use additional phase-estimation, discrete-log, order-finding and hidden-structure examples not copied from the course.                                 |
| Cross-course validation         | Compare against earlier QFT/QPE/Shor evidence and future advanced algorithms modules.                                                                  |
| Benchmark                       | Phase-estimation error vs shots/repetitions; peak width vs $t$ ; convergence iterations; circuit depth;                                                |

|                     |                                                                                                                                                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | controlled-U cost; classical estimator time; retry rate.                                                                                                                                                                 |
| Acceptance criteria | Correct toy-example outputs; bounded estimation error; expected $O(1/t)$ peak-width behavior where applicable; stable adaptive convergence; verified downstream classical result; reproducible telemetry and provenance. |

## 18. Final Course Summary

**Major concepts:** Abelian-group QFT, characters, roots of unity, QFT over  $\mathbb{Z}_N$ , algorithm models, HSP, Abelian/non-Abelian HSP, discrete log, Grover collision search, eigenstates/eigenphases, phase kickback, adaptive estimation and feedback.

**Important algorithms:** Quantum Fourier Transform, Abelian HSP, discrete-log HSP formulation, Grover collision search, Kitaev phase estimation and feedback-based quantum factoring.

**Important problem-solving patterns:** Hidden structure  $\rightarrow$  Fourier representation  $\rightarrow$  measurement; quantum probe  $\rightarrow$  interference  $\rightarrow$  estimation; classical precomputation  $\leftrightarrow$  quantum search trade-off; measurement  $\rightarrow$  likelihood update  $\rightarrow$  adaptive control.

**Important solved questions:** Generalized Bernstein–Vazirani mod 3; collision/hash search; discrete log; Hidden Parabola; quantum factoring by feedback repetition; convergence and adaptive estimator.

**Important notebooks:** None supplied or inspected.

**Primitive candidates:** Adaptive Quantum Spectral Estimation (AQSE); Adaptive Quantum Loop; supporting QFT/character, HSP, phase-estimation and hybrid-search primitives.

**Enterprise Library contributions:** Evidence-backed advanced-algorithm models and candidate adaptive-estimation pattern; not yet a promoted production primitive.

**QAI architecture contributions:** Hybrid Runtime; QAI Control Plane feedback; estimator/measurement telemetry; Capability Registry; Resource Registry; convergence-aware workflow metadata; QAI Labs validation path.

**Implementation opportunities:** Simulator-backed QFT/HSP library; AQSE reference implementation; adaptive phase-estimation controller; hidden-structure solver framework; hybrid collision-search optimizer.

**Evidence gaps:** No executable notebook/QPU evidence; full course URLs/hierarchy not supplied; independent validation and production engineering remain outstanding.

## 19. Extraction Status

| Area                 | Status                           | Notes                                                                                                                |
|----------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Course structure     | Partial / module-level           | SU1 identified; full course hierarchy not independently supplied.                                                    |
| Modules / subunits   | Completed for reviewed evidence  | SU1 Advanced Quantum Algorithms captured.                                                                            |
| Lectures             | Completed for reviewed evidence  | Eight concept areas reviewed in supplied screenshots.                                                                |
| Problem sets         | Completed for reviewed evidence  | Generalized Bernstein–Vazirani, collision search, discrete log, Hidden Parabola, factoring feedback and convergence. |
| Solved questions     | Completed for supplied evidence  | Accepted answers and displayed solutions captured; parser issue explicitly recorded.                                 |
| Notebooks            | Not supplied                     | No executable notebook inspected.                                                                                    |
| Algorithms           | Completed                        | QFT, HSP, discrete log, Grover collision search, phase estimation and adaptive feedback.                             |
| Mathematical models  | Completed                        | Characters, roots of unity, Fourier distributions, eigenphases and feedback likelihood model.                        |
| Patterns             | Completed — candidate            | Probe $\rightarrow$ Measure $\rightarrow$ Estimate $\rightarrow$ Adapt pattern extracted.                            |
| Primitive candidates | Candidate                        | AQSE / adaptive quantum loop; validation required.                                                                   |
| QAI mapping          | Completed — interpretation layer | Architecture mapping separated from MIT evidence.                                                                    |
| Cross-course mapping | Initial                          | Convergence with earlier QFT/QPE/Shor evidence identified; broader course-level mapping pending.                     |

|              |           |                                                                                      |
|--------------|-----------|--------------------------------------------------------------------------------------|
| Validation   | Defined   | Simulator, independent examples, hardware and cross-course validation plan recorded. |
| Final review | Completed | SU1 evidence package completed from supplied study-session material.                 |

20. Extraction Rules

- Preserve the actual MIT OLL hierarchy and terminology.
- Keep original problem statements and solved questions associated.
- Treat solved questions as first-class evidence.
- Treat notebooks and independent implementations as implementation evidence; none was supplied here.
- Do not infer production maturity from educational material.
- Separate MIT source evidence from QAI interpretation.
- Promote recurring patterns to primitive candidates only when evidence supports them.
- Promote primitives toward the Enterprise Library only after independent validation.
- Use cross-course recurrence to strengthen, not replace, validation.
- Record parser/platform issues separately from mathematical correctness.
- Master rule: PRESERVE → EXTRACT → ANALYZE → GENERALIZE → VALIDATE → MAP → PROMOTE.

End State / Next Evidence Package

Module status: COMPLETE

Course status: Part 3 in progress — SU1 completed.

Template status: Standard extraction record populated.

**QAI stack status:** Advanced algorithm concepts, hidden-structure models, phase-estimation mathematics, adaptive feedback pattern, AQSE candidate, hybrid execution mappings and validation plan captured.

**Next step:** Proceed to SU2 of the course and repeat the same evidence → pattern → primitive → QAI mapping process, then consolidate the Part 3 course after its subunits are complete.

Bhadale IT Hub | MIT OLL QAI Learning Extraction

QAI Learning Record | SU2

SU2: Hidden Subgroup Algorithms

Standard Course Extraction Template — Completed Subunit / Evidence Package

**Evidence boundary:** MIT OLL learning is preserved as source evidence. VirtualQubit, QAI Control Plane, validation, workload-allocation and related QAI mappings are study-session architectural interpretations/research candidates, not MIT implementation claims.

1. Course Identity

| Field                         | Captured information                                            |
|-------------------------------|-----------------------------------------------------------------|
| Course title                  | Not independently supplied in the current SU2 screenshots.      |
| MITx / course ID              | Not independently supplied in the current SU2 evidence package. |
| Term / course version         | Not supplied in the inspected evidence.                         |
| MIT Open Learning Library URL | Not supplied in the inspected evidence.                         |
| Primary domain                | Hidden subgroup algorithms, non-Abelian Fourier analysis,       |

|                   |                                                                                                                                                  |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | quantum/classical speedups, communication complexity, amplitude amplification, Hamiltonian simulation and entropy.                               |
| Subunit           | SU2 — Hidden subgroup algorithms                                                                                                                 |
| Purpose           | Capture the supplied SU2 lecture/problem-set learning and extract reusable mathematical, algorithmic, validation and QAI orchestration patterns. |
| Source material   | User-provided MIT OLL screenshots, question text, accepted graded results, displayed course solutions and study-session reasoning.               |
| Extraction status | Completed for the SU2 evidence reviewed in this study session.                                                                                   |

## 2. Course Fidelity and Source Structure

- MIT terminology and the course mathematical framing are preserved at the source-evidence layer.
- Lecture/concept questions and problem-set evidence are kept separate.
- Solved questions are first-class evidence; visible green checks / accepted scores were used as confirmation.
- A parser/input issue occurred on a ket/state entry; the displayed course solution was used to resolve the intended educational result.
- Some ungraded convergence exercises had greyed-out submission controls and official solutions; these are retained as learning evidence, not graded failures.
- No executable notebook/IPYNB, independent simulator, hardware/QPU experiment or external benchmark was supplied.
- QAI architecture interpretations are explicitly separated from MIT evidence.

## 3. Course / Subunit Map

| Item                                        | Type                     | Evidence / focus                                                                  | Status                             |
|---------------------------------------------|--------------------------|-----------------------------------------------------------------------------------|------------------------------------|
| Dihedral HSP problem                        | Lecture / concept        | D_N structure; group size; state preparation/discarded register                   | Completed                          |
| Dihedral HSP algorithm                      | Lecture / concept        | Repeated $ \psi_k\rangle$ samples; subgroup structure                             | Completed                          |
| Shortest vector problem (SVP)               | Lecture / concept        | Lattice and Dihedral-HSP connection                                               | Completed                          |
| Non-Abelian QFT                             | Lecture / concept        | Irreps; unitary representations; commuting matrices                               | Completed                          |
| Polynomial speedups                         | Lecture / concept        | OR lower bound; AND-OR quantum query speedup                                      | Completed                          |
| Amplitude amplification                     | Lecture / concept        | Grover generalization; $2 \times 2$ rotation blocks; boosting                     | Completed                          |
| Application of amplitude amplification      | Lecture / concept        | Rejection sampling; quantum Gibbs sampling                                        | Completed                          |
| Hamiltonian simulation                      | Lecture / concept        | Norm dependence; LCU representation; simulation assumptions                       | Completed                          |
| Distributed Deutsch–Jozsa                   | Problem / worked content | Quantum communication complexity                                                  | Completed                          |
| Communication complexity II — Inner product | Problem set              | Exact inner product; Hadamard; communication lower bound                          | Studied; one parser issue          |
| Communication complexity III — Intersection | Problem set              | Distributed Grover/oracle construction                                            | Completed                          |
| Dihedral HSP — decomposition                | Problem set              | Tensor-product irreps; Clebsch–Gordan decomposition                               | Completed; answers confirmed right |
| Dihedral HSP — one-dimensional irreps       | Problem set              | $\sigma_{\text{triv}}$ , $\sigma_{\text{sign}}$ and $ \psi_{\{k \neq 0\}}\rangle$ | Completed                          |
| Dihedral HSP and QFT                        | Problem set              | Post-QFT state and                                                                | Completed; graded answers          |

|                                  |          |                                |               |
|----------------------------------|----------|--------------------------------|---------------|
|                                  |          | measurement procedure          | accepted      |
| Classical and quantum entropy I  | Exercise | Classical/conditional entropy  | Completed 5/5 |
| Classical and quantum entropy II | Exercise | Von Neumann entropy of a qubit | Completed 1/1 |

## 4. Module / Subunit Record

### 4.1 Learning objectives

- Understand  $D_N$ , its non-Abelian structure and irreducible representations.
- Understand the Dihedral Hidden Subgroup Problem and repeated quantum sampling.
- Understand one-dimensional and two-dimensional irreps and QFT measurement labels.
- Understand tensor-product representation decomposition through a Clebsch–Gordan transformation.
- Connect Dihedral HSP with SVP/lattice structure as presented in the course.
- Understand non-Abelian QFT through irreducible representations.
- Understand quantum query and communication speedups.
- Understand amplitude amplification as a reusable probability-boosting primitive.
- Understand the course framing of Hamiltonian simulation, operator norms and approximation assumptions.
- Relate classical entropy, conditional entropy and von Neumann entropy to uncertainty and state characterization.

### 4.2 Important concepts

- Dihedral group  $D_N$ ; non-Abelian group structure; hidden subgroup  $H$  and hidden shift  $y$ .
- One-dimensional/two-dimensional irreps; representation labels; tensor products.
- Clebsch–Gordan decomposition; non-Abelian QFT.
- Quantum query complexity; distributed quantum communication.
- Grover search; amplitude amplification; rejection/Gibbs sampling.
- Hamiltonian simulation; Trotter/product-formula error; linear-combination-of-unitaries.
- Classical entropy; conditional entropy; density matrices; Bloch vector; von Neumann entropy.

## 5. Lecture / Content Record

| Topic                   | Core learning                                                                                                                                   | QAI interpretation                                 |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Dihedral HSP            | $D_N$ has $2N$ elements; hidden-subgroup information is extracted from quantum states and representation labels.                                | Hidden-structure solver / group capability profile |
| Dihedral HSP algorithm  | Repeated copies of $ \psi_k\rangle$ are used to obtain useful $k$ values.                                                                       | Sampling + inference orchestration                 |
| SVP                     | Lattice points are infinite; course connects Dihedral HSP and SVP.                                                                              | Lattice/hidden-structure capability metadata       |
| Non-Abelian QFT         | Finite-group representations provide the Fourier basis; irreducibility constrains commuting matrices.                                           | Representation registry / mathematical validation  |
| Polynomial speedups     | Classical OR needs $\Omega(n)$ queries; AND-OR trees admit $O(\sqrt{n})$ quantum query complexity in the supplied framing.                      | Quantum/classical workload selection               |
| Amplitude amplification | Grover with multiple marked items is a special case; reflection product acts through $2 \times 2$ rotations; randomized success can be boosted. | Reusable amplification primitive                   |
| Applications            | Rejection sampling gives a random point in a shape; quantum Gibbs sampling requires eigenvalue estimates in the supplied framing.               | Sampling / estimator control                       |
| Hamiltonian simulation  | Runtime depends on $\ H\ $ ; improved method assumes an LCU representation in the supplied material.                                            | Simulation capability and backend selection        |

## 6. Problem Set / Solved Learning Record

### Communication complexity — Inner product

Exact inner product leads to an  $n$ -qubit communication implication in the supplied reasoning. One ket entry encountered a parser/evaluator issue; the  $m=n$  result was accepted.

### Communication complexity — Intersection

Distributed Grover search implements an oracle involving both Alice and Bob; the supplied problem states  $O(\sqrt{n} \log_2 n)$  quantum communication.

### Dihedral HSP — decomposition

The accepted decomposition is  $\sigma_{\{j\} \otimes k} \simeq \sigma_{\{j+k\}} \oplus \sigma_{\{j-k\}}$ , hence  $l=j$  and  $m=k$ . The matrix answers were confirmed right.

### Dihedral HSP — one-dimensional irreps

The exercise asks for  $\sigma_{\text{triv}}/\sigma_{\text{sign}}$  probabilities and  $|\psi_{\{k \neq 0\}}\rangle$ . The course solution for the state is  $(|0\rangle + |1\rangle)/\sqrt{2}$ ; this course convention is retained.

### Dihedral HSP and QFT

Post-QFT two-qubit state and the measurement procedure were solved; displayed graded answers were accepted.

### Classical and quantum entropy I

Baseball example:  $P(\text{win})=0.344$ ,  $P(\text{lose})=0.344$ ,  $P(\text{tie})=0.313$ ;  $H(R)=1.59$  bits; conditional entropy shown as 0.54 bits.

### Classical and quantum entropy II

For  $\rho = (I + \vec{r} \cdot \vec{\sigma})/2$ , the accepted answer is  $S(\rho) = H((1+r)/2)$ .

## 7. Key Mathematical Models

| Model                            | Expression / structure                                                                               | Role                           |
|----------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------|
| Dihedral group                   | $ D_N =2N$                                                                                           | Non-Abelian HSP                |
| 2D irrep                         | $\sigma_j(s) = \text{diag}(e^{i(2\pi j/N)}, e^{-i(2\pi j/N)})$ ;<br>$\sigma_j(r) = [[0, 1], [1, 0]]$ | Fourier/representation basis   |
| Tensor decomposition             | $\sigma_j \otimes \sigma_k \simeq \sigma_{\{j+k\}} \oplus \sigma_{\{j-k\}}$                          | Clebsch–Gordan transformation  |
| QFT over a finite group          | Fourier transform defined using irreducible representations and dimension normalization              | Hidden-structure extraction    |
| Grover / amplitude amplification | $O(\sqrt{N/M})$ queries for $N$ items with $M$ marked items in the supplied framing                  | Search / probability boosting  |
| Distributed intersection         | $O(\sqrt{n} \log_2 n)$ quantum communication in the supplied problem statement                       | Communication advantage        |
| Qubit entropy                    | $S(\rho) = H((1+r)/2)$ , $r =  \vec{r} $                                                             | Quantum information diagnostic |
| Hamiltonian simulation           | Runtime depends on $\ H\ $ ; supplied improved algorithm assumes LCU representation                  | Simulation/resource model      |

## 8. Reusable Algorithmic Patterns

- Hidden-Structure: encode subgroup information  $\rightarrow$  Fourier/representation transform  $\rightarrow$  measure  $\rightarrow$  infer hidden parameter.
- Representation-Decomposition: tensor-product irrep  $\rightarrow$  Clebsch–Gordan decomposition  $\rightarrow$  smaller blocks  $\rightarrow$  structured inference.
- Quantum-Sampling-to-Inference: repeated quantum samples  $\rightarrow$  distribution estimate  $\rightarrow$  classical reconstruction.
- Distributed-Oracle: split oracle knowledge across parties  $\rightarrow$  quantum communication  $\rightarrow$  global search.
- Amplitude-Boosting: success amplitude  $\rightarrow$  reflection/amplification  $\rightarrow$  rotation toward marked subspace  $\rightarrow$  measurement.
- Simulation-Abstraction: Hamiltonian model  $\rightarrow$  representation/decomposition  $\rightarrow$  approximate evolution  $\rightarrow$  error/resource validation.

- Information-Diagnostic: probability distribution/density matrix → entropy → uncertainty/mixedness measure.
- Hybrid Execution: quantum transformation produces structured evidence; classical computation performs inference, control and verification.

## 9. QAI Data Structure / Primitive Candidates

- QAI\_GroupProfile — group, order, generators, relations, Abelian/non-Abelian status and supported irreps.
- QAI\_IrrepProfile — label, dimension, representation matrices, equivalence and decomposition rules.
- QAI\_HSPProblem — group, oracle, hidden subgroup promise, query model, measurement model and inference target.
- QAI\_FourierTransformProfile — group, representation basis, normalization, forward/inverse transform and backend capability.
- QAI\_QuantumState / VirtualQubit — logical state/resource context, amplitudes/phases where representable, mapping and lifecycle metadata.
- QAI\_MeasurementRecord — basis, outcomes, shots, estimates, confidence and provenance.
- QAI\_Constraint — hidden parameter and equations/relations derived from measurements.
- QAI\_SamplingProfile — target distribution, success probability, amplification and confidence.
- QAI\_SimulationProfile — Hamiltonian representation, norm, approximation method, error budget and resources.
- QAI\_EntropyProfile — entropy type, input state/distribution and diagnostic result.

## 10. QAI Stack Mapping

| Layer                     | Potential mapping                                                                                                                |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| QAI Language              | Symbolic groups, irreps, states, operators, HSP promises, measurements and constraints.                                          |
| QAI Algorithm Layer       | Reusable HSP, QFT, Grover/amplitude-amplification, communication and simulation templates.                                       |
| QAI Compiler              | Representation-aware decomposition, tensor/irrep decomposition, circuit optimization and logical-to-native mapping.              |
| QAI Runtime               | Coordinate quantum transforms, measurements, sampling and classical inference.                                                   |
| QAI Control Plane         | Backend/algorithm selection, shots, measurement handling, confidence thresholds, retries and post-processing.                    |
| VirtualQubit              | Logical/virtual quantum-resource abstraction and execution context; not a replacement for the physical qubit.                    |
| Validation / Sanity Layer | Normalization, probabilities, representation dimensions, unitarity assumptions, entropy bounds and expected-distribution checks. |
| Capability Registry       | Supported groups, irreps, QFT variants, amplification methods and simulation methods.                                            |
| Resource Registry         | Logical, virtual and physical quantum resources plus classical compute resources.                                                |

## 11. Engineering Insights

- Hidden-subgroup algorithms are naturally hybrid: quantum interference/representation transforms generate structured evidence; classical computation interprets it.
- Representation basis is a first-class algorithmic choice; QAI should know group/irrep capabilities before execution.
- Tensor-product decomposition can expose smaller computational blocks and inform compiler/resource decisions.
- Measurement outcomes should be treated as evidence with provenance.
- Repeated sampling requires explicit shot, confidence and convergence metadata.
- Communication complexity can be a QAI workload/resource constraint.
- Amplitude amplification should be exposed as a reusable primitive with success-probability and iteration assumptions.
- Hamiltonian simulation requires explicit model, norm, approximation and error-budget metadata.
- Entropy can serve as an information/state diagnostic, but production thresholds require independent validation.
- VirtualQubit and Control Plane are best treated as abstraction/control layers over logical algorithms and physical execution resources.

## 12. QAI Knowledge-Graph / Taxonomy Candidates

- Quantum Algorithms → Hidden-Structure Algorithms → Hidden Subgroup Problem → Dihedral HSP
- Quantum Algorithms → Fourier Algorithms → Non-Abelian QFT
- Quantum Mathematics → Representation Theory → Irreducible Representations → Clebsch–Gordan Decomposition
- Quantum Algorithms → Search → Grover → Amplitude Amplification
- Quantum Algorithms → Simulation → Hamiltonian Simulation
- Quantum Communication → Communication Complexity → Distributed Search / Inner Product
- Quantum Information → Entropy → Von Neumann Entropy
- QAI Compiler → Representation-Aware Decomposition
- QAI Control Plane → Sampling / Measurement / Confidence Management
- QAI Virtual Resource → VirtualQubit → Logical-to-Physical Mapping
- QAI Validation → Mathematical Sanity → State / Probability / Operator / Entropy Checks

## 13. Classification and Reuse Assessment

| Classification          | Assessment                    | Result                                                                                                                  |
|-------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Educational knowledge   | Primary                       | Dihedral HSP, non-Abelian QFT, irreps, SVP connection, amplification, simulation, communication complexity and entropy. |
| Mathematical model      | Strong                        | Representation matrices, tensor decomposition, Fourier states, complexity scalings and entropy.                         |
| Algorithm / procedure   | Strong                        | HSP sampling, QFT/irrep measurement, distributed search, amplification and simulation.                                  |
| Pattern                 | Candidate / strong conceptual | Quantum sampling → structured measurement → classical inference; representation-aware execution; hybrid control.        |
| Implementation evidence | Educational only              | MIT OLL graded interface/screenshots; no independent implementation supplied.                                           |
| Reusable capability     | Candidate                     | Independent implementation, benchmarks and validation required.                                                         |
| Production maturity     | Not established               | No production/QPU claim inferred.                                                                                       |

## 14. Validation and Promotion Plan

| Validation area           | Plan                                                                                                                       |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Reference validation      | Reproduce supplied Dihedral HSP matrix/decomposition and entropy examples independently.                                   |
| Representation validation | Verify irrep dimensions, representation relations and tensor-product decompositions.                                       |
| State validation          | Check normalization, probability sums and expected measurement distributions.                                              |
| Algorithm validation      | Implement small Dihedral HSP, distributed intersection and amplitude-amplification examples in a simulator.                |
| Communication validation  | Measure communication against the stated protocol assumptions and input model.                                             |
| Simulation validation     | Compare Hamiltonian simulation approximation with exact small-matrix evolution and quantify error.                         |
| Entropy validation        | Verify classical and von Neumann entropy, including pure/maximally mixed limits.                                           |
| Cross-layer validation    | Verify logical intent, VirtualQubit metadata, backend mapping, measurements and telemetry.                                 |
| Boundary/fault testing    | Inject malformed states, invalid representations, inconsistent metadata and insufficient sampling; confirm reject/escalate |

|                |                                                                                                                                                                   |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | behavior.                                                                                                                                                         |
| IV&V           | Independent review of requirements, assumptions, implementation results and acceptance criteria.                                                                  |
| Promotion gate | Candidate → independent implementation → simulator benchmark → recurrence across courses → hardware validation where relevant → Enterprise Library consideration. |

## 15. Cross-Course Convergence

| Prior learning / asset               | Common pattern                                                                                  | Status                                           |
|--------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------|
| 8.370.2x SU2 — Deutsch–Jozsa / Simon | Hidden structure; quantum execution → measurement → classical inference.                        | Strong conceptual convergence                    |
| 8.370.2x SU3 — QFT / Shor            | QFT, eigenphase/order finding and hidden periodicity connect to HSP.                            | Strong                                           |
| 8.370.2x SU4 — Grover / simulation   | Amplitude amplification and Hamiltonian simulation recur as reusable primitives.                | Strong                                           |
| 8.371.x state / operations / entropy | State validity, operator structure, entropy and measurement diagnostics support QAI validation. | Strong conceptual convergence                    |
| QAI VirtualQubit / Control Plane     | Logical state/resource abstraction plus classical orchestration and measurement control.        | Architectural candidate; recurring pattern       |
| QAI Validation / Sanity Fabric       | Mathematical consistency, probability, normalization and evidence checks.                       | Architectural candidate; requires implementation |

## 16. Evidence Gaps and Boundaries

- Complete MIT course hierarchy, exact SU2 source URL and course version were not independently supplied in the inspected evidence.
- Evidence is based on user-provided screenshots, question text, accepted results and displayed course solutions.
- No executable notebook/IPYNB, independent simulator, hardware/QPU experiment or external benchmark was supplied.
- Course-specific parser conventions and displayed solutions are preserved where they determined accepted answers; they should not be generalized as universal conventions without independent verification.
- VirtualQubit, QAI Control Plane, validation services, registries and workload-allocation mappings are architectural interpretations/research candidates, not MIT implementation claims.
- No production, enterprise-certification or quantum-advantage claim is inferred from this educational evidence.
- SVP/complexity statements are retained as course learning evidence; no new complexity-theory proof is asserted.

## 17. Extraction Status

| Area                    | Status                          | Notes                                                                                     |
|-------------------------|---------------------------------|-------------------------------------------------------------------------------------------|
| Course structure        | Partial / subunit-level         | SU2 evidence captured; no independent full course crawl.                                  |
| Lectures / concepts     | Completed                       | Supplied concept-question groups reviewed.                                                |
| Problem sets            | Completed for supplied evidence | Communication, Dihedral HSP/QFT and entropy evidence reviewed.                            |
| Solved questions        | Completed                       | Accepted results and course-solution evidence captured.                                   |
| Notebooks               | Not supplied                    | No IPYNB/code asset inspected.                                                            |
| Algorithms / procedures | Completed                       | HSP, non-Abelian QFT, distributed search, amplification and simulation patterns captured. |
| Mathematical models     | Completed                       | Irreps, tensor decomposition, Fourier states, complexity scalings and entropy.            |
| Patterns                | Completed / candidate           | Hybrid inference, representation-aware execution and validation patterns.                 |

|                      |                            |                                                                                       |
|----------------------|----------------------------|---------------------------------------------------------------------------------------|
| Primitive candidates | Candidate                  | Independent implementation and validation required.                                   |
| QAI mapping          | Completed / interpretation | Architecture mappings separated from MIT evidence.                                    |
| Cross-course mapping | Initial / strong           | HSP, QFT, Grover, simulation, state/entropy and Control Plane convergence identified. |
| Validation           | Defined                    | Reference, simulator, statistical, cross-layer and IV&V paths identified.             |
| Final review         | Completed                  | SU2 evidence package completed for supplied material.                                 |

## 18. QAI Retention Snapshot

SU2 should be retained as a foundational QAI evidence package for hidden-structure discovery, representation-aware quantum algorithms, quantum/classical communication complexity, amplitude amplification, Hamiltonian simulation and information/entropy diagnostics.

- Hidden Subgroup Problem is retained as a reusable algorithmic family rather than a single algorithm.
- Group/irrep metadata, non-Abelian Fourier transforms and Clebsch–Gordan decomposition are primary mathematical assets.
- Quantum transformation and sampling followed by classical inference and verification is a recurring hybrid pattern.
- VirtualQubit is retained as a logical/resource abstraction and the QAI Control Plane as the coordination layer for backend choice, measurement, sampling, confidence and classical post-processing.
- Validation retains normalization, probability, representation, operator, entropy and provenance checks.
- SU2 reinforces the broader pattern: logical algorithm → quantum execution → measurement → classical inference → validation → feedback/control.

## 19. Extraction Rules Applied

- PRESERVE — retain course terminology, problem statements, mathematical framing and accepted evidence.
- EXTRACT — capture concepts, equations, algorithms, procedures and solved patterns.
- ANALYZE — distinguish source evidence from study-session interpretation.
- GENERALIZE — identify reusable hidden-structure, representation, sampling, communication, simulation and validation patterns.
- VALIDATE — independently test mathematical and implementation behavior before reuse.
- MAP — connect validated patterns to QAI Language, Compiler, Runtime, Control Plane, VirtualQubit and validation services.
- PROMOTE — move only validated and recurring patterns toward reusable QAI Enterprise Library assets.

**MIT OLL Evidence → Pattern → Primitive Candidate → Validation → QAI Enterprise Library**

**Bhadale IT Hub | MIT OLL QAI Learning Extraction**

# Quantum Information Science II, Part 3

## SU3 — Quantum Information Theory

Standard Course Extraction Template — Completed Subunit / Evidence Package

**Evidence boundary:** MIT OLL/course learning is preserved as source evidence. QAI architecture mappings, VirtualQubit, information-aware orchestration, Control Plane, measurement/POVM services and related primitive candidates are study-session architectural interpretations/research candidates, not claims that MIT specifies or validates a production QAI implementation.

**Extraction date:** 2026-08-17

**Extraction status:** COMPLETED for the SU3 evidence reviewed in this study session.

**Evidence source:** User-provided MIT Open Learning Library screenshots, question text, accepted graded results, displayed official solutions and study-session reasoning.

**Notebook / executable asset:** None supplied or inspected for this SU3 package.

## 1. Course Identity

|                                      |                                                                                                                                                                                                                         |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Course title</b>                  | Quantum Information Science II, Part 3 — Advanced quantum algorithms and information theory                                                                                                                             |
| <b>MITx / course ID</b>              | Not independently supplied in the current SU3 screenshots; course identity retained from the Part 3 study-session context.                                                                                              |
| <b>Term / course version</b>         | Not supplied in the inspected SU3 evidence package.                                                                                                                                                                     |
| <b>MIT Open Learning Library URL</b> | Not supplied in the inspected SU3 evidence package.                                                                                                                                                                     |
| <b>Primary domain</b>                | Quantum information theory, classical/quantum entropy, source and channel coding, mutual information, communication capacity, quantum information, quantum channels, random access codes, Holevo information and POVMs. |
| <b>Subunit</b>                       | SU3 — Quantum Information Theory                                                                                                                                                                                        |
| <b>Purpose</b>                       | Capture the supplied SU3 lecture/problem-set evidence and extract reusable information-theoretic, communication, measurement and QAI architecture patterns.                                                             |
| <b>Source material available</b>     | User-provided MIT OLL screenshots, question text, accepted graded answers, displayed official solutions and study-session reasoning.                                                                                    |
| <b>Extraction status</b>             | Completed for the SU3 evidence reviewed in this study session.                                                                                                                                                          |

## 2. Course Fidelity and Source Structure

- MIT terminology and the course's information-theoretic framing are preserved at the source-evidence layer.
- Lecture/concept questions and problem-set evidence are kept separate.
- Solved questions are first-class evidence; visible green checks / accepted scores were used as confirmation.
- The displayed problem-set results included entropy, typical-sequence, quantum random access code and Holevo-theorem exercises.
- A parser/input issue was encountered in quantum-state expression entry during the course work; where a displayed official solution was supplied, the course solution was used to resolve the intended mathematical result.
- No executable notebook, independent simulator, hardware/QPU experiment or external benchmark was supplied for this SU3 package.
- QAI architecture interpretations are explicitly separated from MIT evidence.

## 3. Course / Subunit Map

| Item                                               | Type              | Evidence / result                                           | Status    |
|----------------------------------------------------|-------------------|-------------------------------------------------------------|-----------|
| SU3 — Quantum Information Theory                   | Subunit           | Supplied lecture + problem-set evidence                     | Completed |
| Noiseless coding theorem                           | Lecture / concept | Entropy, compression limit, concavity                       | Completed |
| Typical sequences                                  | Lecture / concept | Typical-set concentration and source coding                 | Completed |
| Noisy coding theorem                               | Lecture / concept | Channel capacity and reliable transmission                  | Completed |
| Channel capacity and mutual information            | Lecture / concept | Correlation, deterministic input and information flow       | Completed |
| Conditional entropy                                | Lecture / concept | Conditioning and information relationships                  | Completed |
| Achievability of reliable communication            | Lecture / concept | Random codebooks and joint typicality                       | Completed |
| Entropic quantities                                | Lecture / concept | Entropy, mutual information, conditional mutual information | Completed |
| Conditional mutual information and data processing | Lecture / concept | Markov chains and information-processing limits             | Completed |

|                                        |                   |                                                              |           |
|----------------------------------------|-------------------|--------------------------------------------------------------|-----------|
| Properties of quantum channels         | Lecture / concept | Von Neumann entropy, Holevo information, entanglement        | Completed |
| Typical sequences — numerical exercise | Problem set       | $H(X)=0.92$ for $p(a)=0.8$ , $p(b)=p(c)=0.1$                 | Completed |
| Typical sequences II                   | Problem set       | $\delta$ -typical probability; accepted result 0.62          | Completed |
| Typical sequences III                  | Problem set       | 98-bit typical-set representation vs 93-bit entropy estimate | Completed |
| Quantum random access codes            | Problem set       | Two-bit/one-qubit retrieval probability 0.854                | Completed |
| Holevo's Theorem I                     | Problem set       | $H(X)=2$ ; $\rho=I/2$ ; Holevo bound 1                       | Completed |
| Holevo's Theorem II                    | Problem set       | Tetrahedral POVM; $I_0=\log_2(4/3)\approx 0.415$ bits        | Completed |

## 4. Module / Subunit Record

|                            |                                                                                                                                                                                                                                                                                                              |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Learning objectives</b> | Understand Shannon entropy, typical sequences, source coding, noisy-channel coding, channel capacity, mutual/conditional mutual information, quantum entropy, quantum channels, random access codes, Holevo information and POVMs.                                                                           |
| <b>Lectures / topics</b>   | Noiseless coding theorem; typical sequences; noisy coding theorem; channel capacity and mutual information; conditional entropy; achievability of reliable communication; entropic quantities; conditional mutual information/data processing; properties of quantum channels.                               |
| <b>Problem-set focus</b>   | Numerical typical-set analysis; quantum random access codes; Holevo theorem and density-matrix analysis; optimal POVM for a tetrahedrally symmetric ensemble.                                                                                                                                                |
| <b>Important concepts</b>  | Shannon entropy $H(X)$ ; typical set; source coding; channel capacity; mutual information $I(X;Y)$ ; conditional entropy $H(X Y)$ ; conditional mutual information; Markov chains; density matrix $\rho$ ; Von Neumann entropy $S(\rho)$ ; quantum channels; Holevo bound; POVM; quantum random access code. |
| <b>References</b>          | MIT Open Learning Library material supplied in the study session; displayed official solutions; standard MIT OLL QAI extraction template/process.                                                                                                                                                            |

## 5. Lecture / Content Record

| Lecture / item                          | Source learning                                                                                                      | Mathematical idea                                          | QAI relevance                                          |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------|
| Noiseless coding theorem                | Entropy is concave; iid strings have a compression limit tied to entropy.                                            | Entropy as information rate.                               | Information-aware data representation and compression. |
| Typical sequences                       | Most probability mass concentrates on a typical set; typical sequences support asymptotically optimal source coding. | Typical-set size $\approx 2^{(nH(X))}$ .                   | Probability-aware state/data reduction.                |
| Noisy coding theorem                    | Reliable transmission is possible below channel capacity.                                                            | Capacity-constrained reliable communication.               | QAI communication profiles and orchestration.          |
| Channel capacity and mutual information | Mutual information is small for weakly correlated variables; deterministic input gives zero mutual information.      | $I(X;Y)$ quantifies shared information.                    | Information-flow/workload-routing metric.              |
| Conditional entropy                     | Conditioning reduces entropy; relative-entropy relationships affect mutual information.                              | $H(X Y)$ and $I(X;Y)$ .                                    | Context-aware inference and information-flow analysis. |
| Reliable communication                  | Random codebooks and joint typicality support reliable decoding.                                                     | Joint typicality.                                          | Probabilistic communication assurance.                 |
| Entropic quantities                     | Mutual information is zero iff variables are independent;                                                            | Relationships among $H$ , $H $ , $I$ and conditional $I$ . | Information-health diagnostics.                        |

|                                                |                                                                                                                               |                                                                    |                                                        |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------|
|                                                | entropy quantities distinguish dependence and determinism.                                                                    |                                                                    |                                                        |
| Conditional mutual information/data processing | Markov structure implies conditional independence and limits information transfer through processing.                         | $I(X;Y Z)$ , data processing.                                      | Provenance-aware QAI pipelines.                        |
| Properties of quantum channels                 | Quantum compression uses Von Neumann entropy; Holevo information is non-negative; entanglement can increase channel capacity. | $S(\rho)=-\text{Tr}(\rho \log \rho)$ ; quantum/classical boundary. | Quantum-state representation and hybrid communication. |

## 6. Problem Set Record

|                                     |                                                                                                                                   |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Problem set / evidence group</b> | SU3 Quantum Information Theory problem-set questions reviewed in the study session.                                               |
| <b>Question count</b>               | Multiple displayed problem sets and multi-part exercises; exact official total was not independently supplied in the screenshots. |
| <b>Original problem available</b>   | Yes — user supplied screenshots and problem statements.                                                                           |
| <b>Solved questions available</b>   | Yes — accepted graded results and displayed solutions were reviewed.                                                              |
| <b>Notebook / code available</b>    | No executable notebook or code asset supplied.                                                                                    |
| <b>Related lectures</b>             | Entropy/coding; typical sequences; noisy channels; mutual information; quantum channels; Holevo theorem; POVMs.                   |

## 7. Solved Question Extraction

| Question                    | Problem / result                                                                             | Key reasoning                                                   | Validation                                                         |
|-----------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------|
| Typical sequences — entropy | $p(a)=0.8$ , $p(b)=p(c)=0.1$ ; calculate $H(X)$ .                                            | $H(X)=-\sum p(x)\log_2 p(x)$ .                                  | Accepted: 0.92 bits/symbol.                                        |
| Typical sequences II        | $n=100$ , $\delta=0.1$ ; calculate probability of typical set.                               | Use $f(n,n_a)=0.8^{n_a}0.1^{n-n_a}$ and typicality bounds.      | Accepted: 0.62.                                                    |
| Typical sequences III       | Compare typical-set representation with $nH(X)$ .                                            | Typical-set cardinality is exponential in $nH(X)$ .             | Accepted: 98 whole bits vs 93-bit entropy estimate.                |
| Quantum random access codes | Encode multiple classical bits into one qubit and retrieve a selected bit probabilistically. | Projective measurement probability follows state overlap.       | Accepted: $p=0.854$ .                                              |
| Holevo's Theorem I          | Four equally likely pure qubit states: $H(X)$ , $\rho$ and Holevo bound.                     | $H(X)=2$ ; $\rho=I/2$ ; pure-state entropies vanish.            | Accepted: $H(X)=2$ , $\rho=[[1/2,0],[0,1/2]]$ , bound 1.           |
| Holevo's Theorem II         | Construct tetrahedral POVM and calculate accessible mutual information.                      | $M_k= \phi_k\rangle\langle\phi_k /2$ ; symmetry fixes the POVM. | Accepted: $I_0=\log_2(4/3)\approx 0.415$ bits; bound not achieved. |

## 8. Notebook / Computational Asset Record

|                                            |                                                                                               |
|--------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Notebook name</b>                       | None supplied.                                                                                |
| <b>Course / problem-set relationship</b>   | No executable notebook inspected.                                                             |
| <b>Original source</b>                     | User-provided MIT OLL screenshots.                                                            |
| <b>Language / framework / dependencies</b> | Not applicable.                                                                               |
| <b>Input</b>                               | Problem-set parameters and distributions shown in screenshots.                                |
| <b>Output</b>                              | Accepted numerical and symbolic results displayed by MIT OLL.                                 |
| <b>Algorithm / model</b>                   | Entropy, typical-set counting, quantum measurement, POVM and Holevo-information calculations. |
| <b>Execution environment</b>               | MIT OLL assessment interface.                                                                 |
| <b>Validation</b>                          | Course grader / displayed official solution; independent implementation remains future work.  |

## 9. Algorithm / Mathematical Model Record

| Model | Core structure | Role | QAI relevance |
|-------|----------------|------|---------------|
|-------|----------------|------|---------------|

|                                |                                                      |                                              |                                       |
|--------------------------------|------------------------------------------------------|----------------------------------------------|---------------------------------------|
| Shannon entropy                | $H(X) = -\sum p(x) \log_2 p(x)$                      | Classical uncertainty / information rate     | Information profiling                 |
| Typical set                    | $2^{-n(H+\delta)} \leq p(x^n) \leq 2^{-n(H-\delta)}$ | High-probability sequence region             | Compression / reduction               |
| Mutual information             | $I(X;Y) = H(X) - H(X Y)$                             | Shared information                           | Information-flow analysis             |
| Conditional mutual information | $I(X;Y Z)$                                           | Residual dependence after conditioning       | Data-processing analysis              |
| Channel capacity               | $C = \max_{p(x)} I(X;Y)$                             | Maximum reliable communication rate          | Channel-aware routing                 |
| Density matrix                 | $\rho = \sum p_x  \psi_x\rangle\langle\psi_x $       | Quantum ensemble representation              | VirtualQubit state abstraction        |
| Von Neumann entropy            | $S(\rho) = -\text{Tr}(\rho \log_2 \rho)$             | Quantum information content                  | Quantum compression/state uncertainty |
| Holevo quantity                | $\chi = S(\rho) - \sum p_x S(\rho_x)$                | Accessible classical-information upper bound | Quantum/classical boundary            |
| POVM                           | $M_k \geq 0, \sum M_k = I$                           | Generalized quantum measurement              | Measurement planning                  |
| Quantum random access code     | Quantum encoding + selected-basis measurement        | Information encoding vs retrievability       | VirtualQubit/measurement design       |

## 10. Extracted Patterns

| Pattern                                        | Structure / purpose                                                                                              | Potential QAI reuse                                            | Evidence                                                |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------|
| Information-Aware Representation & Compression | Entropy $\rightarrow$ typicality $\rightarrow$ reduced representation $\rightarrow$ reconstruction.              | QAI data fabric, RAG, knowledge representation, digital twins. | High educational evidence; QAI reuse is interpretation. |
| Channel-Aware Reliable Information Transfer    | Source $\rightarrow$ encoder $\rightarrow$ noisy channel $\rightarrow$ decoder, constrained by capacity.         | QAI communication fabric and distributed execution.            | High educational evidence.                              |
| Information-Flow / Data-Processing Analysis    | $X \rightarrow Y \rightarrow Z$ with mutual/conditional information relationships.                               | QAI pipelines, provenance, inference and orchestration.        | Conceptual design candidate.                            |
| Quantum-to-Classical Information Boundary      | Quantum state $\rightarrow$ measurement $\rightarrow$ classical outcome; Holevo bounds accessible information.   | VirtualQubit, measurement service, Control Plane.              | Strong conceptual mapping.                              |
| Measurement-Optimized Information Extraction   | Quantum ensemble $\rightarrow$ POVM $\rightarrow$ classical outcome $\rightarrow$ mutual-information evaluation. | Measurement orchestration and validation.                      | Research/design candidate.                              |

## 11. Primitive Candidates

| Primitive                      | Purpose                                                                                           | Maturity             |
|--------------------------------|---------------------------------------------------------------------------------------------------|----------------------|
| InformationProfile             | Represent entropy, mutual information, conditional entropy and channel metrics.                   | Candidate / research |
| TypicalSetReducer              | Identify high-probability typical regions and compact representations.                            | Candidate / research |
| QAIChannelProfile              | Represent capacity, noise assumptions, rate and reliability targets.                              | Candidate / research |
| QuantumInformationState        | Abstract density-matrix/state information and provenance.                                         | Candidate / research |
| POVMMeasurementPlanner         | Compare measurement strategies using accessible-information objectives.                           | Candidate / research |
| HolevoBoundValidator           | Check proposed classical extraction against a quantum ensemble's Holevo bound.                    | Candidate / research |
| InformationAwareWorkloadRouter | Use information characteristics plus execution capability for classical/quantum/hybrid placement. | Candidate / research |

## 12. QAI Stack Mapping

| QAI layer    | Proposed mapping                          | Evidence classification      |
|--------------|-------------------------------------------|------------------------------|
| QAI Language | Represent entropy, distributions, density | Architectural interpretation |

|                            |                                                                                                                                            |                              |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
|                            | matrices, channels, POVMs and information constraints symbolically.                                                                        |                              |
| QAI Data / Knowledge Layer | Store information profiles, probability models, state metadata and provenance.                                                             | Architectural interpretation |
| QAI Compiler               | Use information and measurement constraints in transformation/representation choices.                                                      | Architectural interpretation |
| QAI Runtime                | Execute classical and quantum transformations, measurements and communication workflows.                                                   | Conceptual mapping           |
| QAI Control Plane          | Monitor information-flow metrics, channel conditions, measurement outcomes and policy thresholds.                                          | Architectural interpretation |
| VirtualQubit               | Expose logical/virtual state abstraction with state and measurement metadata while decoupling application logic from physical realization. | Research/design candidate    |
| Hybrid Runtime             | Coordinate quantum state preparation/measurement with classical coding, estimation and inference.                                          | Conceptual mapping           |
| QAI Intelligence Layer     | Interpret quantum-generated measurements and choose next actions using classical inference.                                                | Architectural interpretation |
| Validation / Assurance     | Check entropy bounds, capacity assumptions, Holevo bounds, measurement probabilities and information-loss constraints.                     | Research/design candidate    |

## 13. Engineering Insights

- Treat information as a managed computational resource, not only as a data payload.
- Do not assume a large state space requires explicit processing of every state; typicality can identify a much smaller high-probability region.
- Separate information encoded in a quantum state from classical information that can actually be extracted by measurement.
- Use mutual information and conditional mutual information as candidate observability metrics for information flow.
- Treat channel capacity and noise characteristics as first-class inputs to distributed QAI communication decisions.
- Keep quantum-state representation, measurement strategy and classical interpretation as distinct layers.
- VirtualQubit is best treated as an abstraction boundary rather than a claim about a specific physical qubit implementation.
- The QAI Control Plane can conceptually use information metrics for measurement policy, communication policy and workload-placement decisions.
- Production claims require independent implementation and validation.

## 14. QAI Information-Flow Architecture

**Source / State → Information Profile → QAI Control Plane → Workload Selection → Execution → Measurement / Decode → Information Validation**

- Classical path: entropy estimation → compression / classical processing → reconstruction.
- Quantum path: density matrix / VirtualQubit state → quantum processing → POVM / measurement → classical result.
- Hybrid path: quantum state preparation → quantum execution → classical estimation/inference → control update.
- Control path: information metrics + channel profile + backend capability + validation policy → processor/workload selection.

## 15. Cross-Course Convergence / Reuse

- Builds on quantum states, density matrices, quantum operations, measurements and quantum channels.
- Connects to SU2 Hidden Subgroup Algorithms: quantum interference produces information that is interpreted classically.
- Extends QFT/phase-estimation learning with uncertainty, accessible information and measurement concepts.
- Supports QAI Control Plane ideas around measurement handling, feedback, validation and hybrid execution.

- Strengthens the VirtualQubit concept by placing quantum-state representation and measurement inside an information-theoretic boundary.
- Provides foundations for QAI communication, distributed execution, RAG/knowledge processing and information-aware workload allocation.

## 16. Validation / Evidence

- All displayed lecture concept questions shown in the study session were accepted.
- Typical-sequence numerical exercise:  $H(X)=0.92$  accepted.
- Typical sequences II: probability 0.62 accepted.
- Typical sequences III: 98-bit representation and 93-bit entropy comparison accepted.
- Quantum random access code:  $p=0.854$  accepted.
- Holevo's Theorem I:  $H(X)=2$ ,  $p=1/2$  and Holevo bound 1 accepted.
- Holevo's Theorem II: tetrahedral POVM and  $I_0 \approx 0.415$  bits accepted.

## 17. Evidence Gaps and Boundaries

- Exact MITx course ID and official URL were not independently supplied in the current SU3 screenshots; course identity is retained from the Part 3 study context.
- No executable notebook, simulator implementation, QPU run or hardware/network experiment was supplied.
- QAI primitives and architecture mappings are design candidates derived from the learning, not MIT-prescribed implementations.
- Typical-set and coding results are educational examples and should not be generalized without matching source/channel assumptions.
- Random-access-code probabilities apply to the specific encoding/measurement constructions shown.
- The Holevo bound is an upper bound; the tetrahedral POVM example achieves less than the bound.
- No production maturity, hardware scalability, security certification or enterprise performance is inferred from this material.

## 18. Future Reading / Course Reference

- MIT Open Learning Library — Quantum Information Science II, Part 3 — Advanced quantum algorithms and information theory; SU3: Quantum Information Theory.
- Use subsequent quantum-information and communication material to validate information-aware orchestration, quantum networking, coding and measurement candidates.
- Cross-reference completed QAI extraction records for quantum states/density matrices, quantum operations/channels, hidden subgroup algorithms and advanced quantum algorithms.

## 19. Module Summary

**SU3 establishes the information-theoretic layer needed to reason about what a QAI system stores, transmits, transforms and can actually extract from quantum states.** Shannon entropy and typical sequences establish compression and probability-concentration principles; mutual information, conditional entropy and channel capacity establish information-flow and reliable-communication limits; density matrices and Von Neumann entropy extend these ideas to quantum states; quantum random access codes illustrate information encoding versus retrievability; and Holevo's theorem plus POVMs establish the boundary between quantum information and accessible classical information.

**Strongest QAI architectural learning:** the QAI stack should optimize the flow, representation, transformation and extraction of information—not simply maximize quantum hardware utilization. This supports an information-aware QAI Control Plane, VirtualQubit state abstraction, hybrid runtime, measurement services and future workload-allocation mechanisms.

## 20. Status

|               |                                                                |
|---------------|----------------------------------------------------------------|
| Module status | COMPLETE                                                       |
| Course status | Quantum Information Science II, Part 3 — SU3 completed; course |

|                         |                                                                                                                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | remains in progress.                                                                                                                                                                 |
| <b>Template status</b>  | Standard extraction record populated.                                                                                                                                                |
| <b>Evidence status</b>  | Lecture questions and displayed problem-set questions reviewed with accepted graded results / official solutions.                                                                    |
| <b>QAI stack status</b> | Core concepts, mathematical models, recurring patterns, candidate primitives, Control Plane mappings, VirtualQubit relevance, validation boundaries and cross-course reuse captured. |

## 21. Master Extraction Rule

**PRESERVE → EXTRACT → ANALYZE → GENERALIZE → VALIDATE → MAP → PROMOTE**

SU3 is retained both as a solved-course knowledge set and as a source of candidate reusable QAI information-processing patterns. Promotion into the QAI Enterprise Library remains gated by independent validation.

## Appendix A — Candidate QAI Information Primitive Family

- Entropy Analyzer
- Typical Set Analyzer / Reducer
- Information Profile
- Mutual Information Monitor
- Conditional Information Analyzer
- Channel Capacity Profiler
- Quantum State Information Profiler
- Von Neumann Entropy Analyzer
- Quantum Random Access Code Evaluator
- Holevo Bound Validator
- POVM Measurement Planner
- Information-Aware Workload Selector
- Hybrid Information Flow Monitor
- QAI Communication Health Profile

## Appendix B — QAI Information-Aware Execution Pattern

**Problem / Data → Information Profile → State / Channel Analysis → Classical / Quantum / Hybrid Selection → Execution → Measurement / Decode → Information Validation → Telemetry / Provenance**

## End of SU3 Knowledge Capture

*MIT Quantum Information Science II, Part 3 | SU3 — Quantum Information Theory*